



ARTIFICIAL INTELLIGENCE

Weekly Lecture Schedule • Fall 2026

Bachelor of Science in Information Technology • Search, Knowledge and Reasoning • every example drawn from IT project management

CREDIT HOURS 3 (2–3): 2 lecture h + 3 lab h a week	DURATION 16 teaching weeks; midterm Week 8, final after Week 16	PREREQUISITE Object Oriented Programming (HEC 2023); Semester 3	CONTENTS 22 chapters in 6 parts, each with a runnable Python lab
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📅	📖 Lecture topics	🔧 Laboratory (3 h)	📋 Assessment
1	1 What Artificial Intelligence Is – four definitions on two axes, the “is it AI?” checklist, branches, five waves, the AI/ML/DS boundary 2 Intelligent Agents and Task Environments – percepts and actions, PEAS, the six environment properties, five agent architectures, rationality	Is it AI? a five-point audit; a reflex agent on a ticket queue	—
2	3 Representations and Search in Python – states as values, the Problem interface, three frontiers, dependency graphs, generators and memoisation	The frontier and the graph: the course toolkit	—
3	4 Problem Formulation and Uninformed Search – state space against search tree, the explored set, breadth-first, depth-first, iterative deepening, uniform cost	Task ordering four ways	Quiz 1 (Ch 1–3)
4	5 Informed Search and Heuristics – admissibility and consistency, greedy best-first, A* and its optimality proof, dominance, relaxation	A* over sprint states, two heuristics	Assignment 1 set
5	6 Local Search and Optimisation – the state-space landscape, hill climbing, random restart, simulated annealing, genetic algorithms	Staffing by annealing and by evolution	—
6	7 Constraint Satisfaction – variables, domains and constraints, backtracking, minimum remaining values, forward checking, arc consistency	Sprint scheduling as a CSP	Assignment 1 due; Quiz 2 (Ch 4–6)
7	8 Adversarial Search and Game Playing – game trees, minimax, alpha–beta pruning, move ordering, cutting off search, evaluation functions	Minimax and alpha–beta on a slot negotiation	—
8	9 Planning – the STRIPS representation, progression search, means–ends analysis and the General Problem Solver	A release plan from operators	Midterm (Ch 1–8)
9	10 Knowledge Representation – semantic networks, inheritance, frames and defaults, ontologies, what a representation must support	A semantic network that answers queries	—
10	11 Propositional Logic and Inference – syntax and semantics, entailment, model checking, conjunctive normal form, DPLL, resolution	Three ways to decide a release gate	—
11	12 First-Order Logic and Resolution – quantifiers, unification with the occurs check, skolemisation, resolution refutation 22 The Artificial Intelligence Project – the project brief; choosing and scoping a problem, the plan	Unification and a resolution proof	Project set, proposal due; Assignment 2 set
12	13 Rule-Based Expert Systems – production rules, forward and backward chaining, conflict resolution, explanation, the knowledge bottleneck	A twenty-rule escalation shell that explains itself	Assignment 2 due
13	14 Uncertainty and Decision Making – probability as degree of belief, Bayes’ rule, independence, expected utility, the value of information 15 Bayesian Networks – structure and conditional independence, enumeration, variable elimination, sampling	Go/no-go under uncertainty; diagnosing a defect cause	Quiz 3 (Ch 9–13)
14	16 Fuzzy Logic – membership functions, fuzzy operators, Mamdani inference, defuzzification, fuzzy against probabilistic 17 Learning from Examples – the three paradigms, the perceptron worked by hand, linear separability and the XOR limit	Risk rating by Mamdani inference; the perceptron and its limit	Project experiments frozen
15	18 Natural Language Processing – the symbolic pipeline, context-free grammars, chart parsing, pattern substitution, n-grams 19 Classic AI Systems and How to Analyse Them – the General Problem Solver, Eliza, Student, Macsyma; a six-column analysis framework	Parsing ticket subjects; four systems in one file	—
16	20 Recent Trends and Applications – large language models, tool-using agents, neuro-symbolic systems — as signposts, not syllabus 21 Responsible AI and Its Limits – where bias enters, explainability as an audit artefact, brittleness, computational limits	A tool-using agent, offline; auditing the escalation rules	Project report and presentation

ASSESSMENT WEIGHTS

20% Sessional: quizzes, assignments, project	30% Midterm, Ch 1–8	50% Final, whole course
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I Foundations of Intelligent Systems
IV Knowledge, Logic and Reasoning

II Problem Solving by Search
V Reasoning Under Uncertainty

III Constraints, Games and Plans
VI Learning, Language and Responsible AI

Chapter numbers refer to the Fall 2026 Artificial Intelligence handout (Appendix G, Sixteen-Week Teaching Schedule). Week 8 carries Chapter 9 alongside the midterm on purpose: STRIPS progression search reuses the frontier of Chapters 4 and 5, so the lecture is light in a revision week. This course is deliberately symbolic — Chapter 17 is a signpost to *Machine Learning* (Semester 6), not a unit; Appendix F states every boundary, including the one with *Knowledge Representation and Reasoning*. Weights follow the department’s standard split: Sessional 20%, Mid-term 30%, Final 50%. Set texts (HEC): S. J. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach* (3rd ed., 2010); G. F. Luger and W. A. Stubblefield, *AI Algorithms, Data Structures, and Idioms* (2009); D. Poole and A. Mackworth, *Artificial Intelligence: Foundations of Computational Agents* (2nd ed., 2017).